

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses Poisson $\lambda = 10$ PI	AO1.1a	M1	$V + W$ is $Po(10)$
	Obtains correct probability	AO1.1b	A1	$P(S > 10) = 1 - P(S \leq 10)$ $= 0.417$
(b)	States that model requires independence of purchases from store to store.	AO3.5b	E1	Purchases of printers at <i>Verigood</i> are not independent of those at <i>Winnerprint</i>
				Total 3 marks

Q2.

- (a) (i) $e^{-1.5} \times 1.5^3 / 3!$ M1
- $= 0.126$ A1
- 0.125 to 0.126*
- 2
- (ii) Using $Po(1)$, $P(X < 1) = 1 - P(X \leq 1)$ M1
- $= 1 - 0.7358 = 0.264$
- SC Award M1 only if obtain 0.0902 using $Po(0.5)$*
- A1
- 2
- (iii) Weekdays $Po(7.5)$ weekend $Po(1)$
- Weekdays = 7.5*
- M1
- Total $Po(8.5)$
- A1
- $P(\text{Total} < 10) = P(\text{Total} \leq 9)$
- Applied (0.7764, 0.7166, 0.6530 are evidence)*
- m1

$= 0.653$

A1

4

- (b) Using Total Po from (a)(iii)
 $P(> 15) = 0.0138$, $P(> 16) = 0.0066$
*M1 using their total providing supporting probabilities
 seen OE use of $P(\text{Total} \leq 15 \text{ \& } 16)$*

M1

So needs 16 tubes
CAO Answer alone scores B2

A1

2

- (c) Average rate of failure unlikely to be constant over the course of a day. Very little use of lights over this period.
One mark for any sensible comment

E1

1

[11]

Q3.

(a) (i) $\frac{e^{-3.5} \times 3.5^4}{4!}$

M1

$= 0.189$

AWRT 0.189 Answer only gets B2

©2025 Exam Papers Practice. All rights reserved.

A1

2

- (ii) Using or stating $Po(0.5)$
*An answer of 0.0144, 0.3935, 0.6065, 0.9098 or 0.9856
 implies award of B1 but no further marks*

B1

$P(\geq 2) = 1 - P(\leq 1)$

or

$= 1 - 0.9098$

M1

$= 0.0902$

Accept 0.09

A1

- (iii) Using Po(14)
Sight of 0.1094, 0.1757, 0.9235, 0.9521

B1

$$P(\leq 19) - P(\leq 10) = 0.9235 - 0.1757$$

Allow 0.8752 - 0.1185
or 0.9573 - 0.2517 for M1

M1

$$= 0.7478$$

AWFW 0.747 to 0.748

A1

3

- (b) **GRBs / explosions / events / etc** will be random and / or independent
GRBs / etc short in comparison to observation period (non-overlapping)
For any valid point

E1

1

[9]

Q4.

(a) $\lambda = 6 \times 2.5 = \underline{15}$
CAO

B1

$$P(W \leq 18) = \underline{0.819 \text{ to } 0.82}$$

AWFW (0.8195)

B1

2

- (b) $P(W > w) \leq 0.05$ $P(W \leq w) \geq 0.95$
Implied by a value of 21, 22 or 23

M1

$$w = \underline{22}$$

CAO

A1

2

[4]

Q5.

(a) (i) Poisson

B1

1

(ii) $E(3X - 1) = 3\lambda - 1$

B1

$\text{Var}(3X - 1) = 9\lambda$
oe (allow $3^2\lambda$)

B1

2

(iii) $P(X = x + 1) = \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!}$

B1

$$\begin{aligned}
 P(X = x + 1) &= \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!} \\
 &= \frac{e^{-\lambda} \times \lambda^x \times \lambda}{(x+1)x!} \\
 &= \frac{\lambda}{x+1} \times \frac{e^{-\lambda} \times \lambda^x}{x!} \\
 &\quad \text{dep(B1)}
 \end{aligned}$$

Mdep1

$$= \frac{\lambda}{x+1} P(X = x)$$

AG

©2025 Exam Papers Practice. All rights reserved.

Adep1

3

(b) (i) $\lambda_{\text{car}} = 500$ / hour

$\lambda_{\text{coach}} = 10$ / hour

$\Rightarrow \lambda_{\text{vehicle}} = 510$ / hour = 8.5 / min
for 8.5 stated / used

B1

$P(V \geq 10) = 1 - 0.6530$

M1

= 0.347

A1

special case:

$$\lambda = 10 \Rightarrow B1M0A0$$

$$B1 \Rightarrow 1 - 0.458 \text{ or } 0.542$$

3

(ii) $\mu_{\text{car}} = 836/\text{hour}$

$$\mu_{\text{coach}} = 22/\text{hour}$$

$$\Rightarrow \mu_{\text{vehicle}} = 858 / \text{hour} = 14.3 / \text{min}$$

for 14.3 stated / used

B1

$$P(V \leq 3) = P(V = 0, 1, 2, 3)$$

$$= \begin{cases} e^{-14.3} \left[1 + \frac{14.3}{1} + \frac{14.3^2}{2} + \frac{14.3^3}{6} \right] \\ e^{-14.3} \times 604.91283 \end{cases}$$

= 0.0003726 to 0.000373

*All 4 terms required for **any** $\lambda > 0$*

M0 for use of normal approximation

= 0.00037 (2sf)

dep M1

M1

Adep1

3

EXAM PAPERS PRACTICE [12]

©2025 Exam Papers Practice. All rights reserved.

Q6.

(a) (i) $P(X \geq 9) = 1 - P(X \leq 8)$

$$1 - 0.6530 = 0.347 \text{ (B1)}$$

$$= 1 - 0.5231$$

$$= 0.4769$$

awfw 0.476 and 0.477

B2,1

2

(ii) $P(5 < X < 10) = P(X \leq 9) - P(X \leq 5)$

$$= 0.653 - 0.1496$$

$$= 0.5034$$

awfw 0.503 to 0.504

$$0.7634 - 0.1496 = 0.613 \text{ to } 0.614 \text{ (B2)}$$

$$0.6530 - 0.2562 = 0.397 \text{ to } 0.398 \text{ (B2)}$$

$$0.7634 - 0.2562 = 0.507 \text{ to } 0.508 \text{ (B1)}$$

$$\alpha = 0.1496 \text{ or } 0.653 - \alpha \text{ (B1) iff } 0 < p < 1$$

B3,2,1

3

(b) $P(Y < 2) = P(Y \leq 1) = P(Y = 0 \text{ or } Y = 1)$
 $0.8 \text{ to } 0.81 \text{ (B1)}$

$$= e^{-1.5} + e^{-1.5} \times 1.5$$

(both)

M1

$$[0.2231 + 0.3347]$$

$$= 0.5578254$$

$$= 0.558$$

awfw 0.557 to 0.56

A1

2

(c) (i) $\lambda = 8.5 + 1.5 = 10$
 Allow $P(10)$ or $Po(10)$

B1

1

(ii) $P(T > 16) = 1 - P(T \leq 16)$
 $= 1 - 0.9730$

M1

$$= 0.027$$

A1

2

(iii) $p = {}^3C_2 0.027^2 \times 0.973$
 for either term correct

M1

$$+ 0.027^3$$

for addition of the two correct terms

M1

$$= 0.002128 + 0.00001968$$

$$= 0.0021 \text{ [4 dp]}$$

0.0021 or 0.0022 [iff M1M1 (+ 4dp)]

A1

3

Alternative:

$$p = 1 - P(X \leq 1)$$

$$P(X = 0) + P(X = 1)$$

$$= 0.973^3 + 3 \times 0.973^2 \times 0.027 = 0.921167 + 0.076685$$

for either term correct

(M1)

$$p = 1 - 0.99785$$

for 1 - [sum of two correct terms]

(M1)

$$= 0.0021$$

0.0021 or 0.0022 [iff M1M1 (+ 4dp)]

(A1)

(3)

[13]

Q7.

- (a) (i) $X \sim P_0(0.06)$
 $P(X \leq 1) = 0.8781$
Awrt 0.878

B1

1

- (ii) For matches:
 The number of run outs: $Y \sim P_0(0.15)$

$$P(Y \geq 1) = 1 - P(Y = 0)$$

$$= 1 - e^{-0.15}$$

$$= 1 - 0.8607$$

©2025 Exam Papers Practice. All rights reserved.

must use $P_0(0.15)$.

M1

$$= 0.1393$$

awrt 0.139

A1

$$P(X \leq 1 \text{ and } Y \geq 1) = 0.8781 \times 0.1393$$

their (a)(i) $\times$ their $P(Y \geq 1)$

M1

$$= 0.122$$

awrt

A1

4

- (b) X and Y are independent.

Number of catches and run outs
independent

B1

1

- (c) (i) For Season:

$$S \sim \text{Po}(9.6)$$

*Use of $\lambda = 9.6$ in correct
Poisson expression*

M1

$$P(S = 10) = \frac{e^{-9.6} \times 9.6^{10}}{10!}$$

$$= 0.124$$

A1

2

- (ii) $T \sim \text{Po}(9.6 + 2.4) = \text{Po}(12)$

$\text{Po}(12)$ used or seen

B1

$$P(T \geq 15) = 1 - P(T \leq 14)$$

$$= 1 - 0.7720$$

$$= 0.228$$

$$(1 - 0.8444 = 0.155 \text{ to } 0.156)$$

©2025 Exam Papers Practice. All rights reserved.

B2,1

3

[11]

Q8.

- (a) (i) $X \sim \text{Po}(13)$

Both Poisson and $\lambda = 13$

B1

1

- (ii) $P(X = 20) = P(X \leq 20) - P(X \leq 19)$

$$= 0.975(0) - 0.957(3)$$

[allow 0.975 – 0.957]

Must use $\lambda = 13$ otherwise MOA0

M1

$= 0.0177$ (3sf)

AWFW 0.0176 to 0.018

$$\text{or } P(X = 20) = \frac{e^{-13} \times 13^{20}}{20!} \quad \begin{array}{l} M1 \\ A1 \end{array}$$

$$= 0.0177 \quad A1$$

A1

2

(iii) $P(6 \leq X \leq 18) = P(X \leq 18) - P(X \leq 5)$
 $= 0.930(2)$

M1

$-(0.0107 \text{ or } 0.0259)$

M1

$= 0.920$ (3sf)

AWFW 0.919 to 0.92

A1

3

- (b) Cars not random
 Cars not independent

Allow (number of) cars not random / not independent

Mean and Variance of cars different / not equal
B1 for any one of these 3 statements
*Must indicate a reference to **cars***

B1

Mean / Average / λ / 2.6

Correct comment about value of $\lambda \neq 2.6$

greater / less / smaller / different / variable / not constant /
 too small / too large

Any combination (one from each group):

Any contextual reason that suggests a change in traffic flow, eg due to:
 rush hour / congestion / traffic jams / accidents / work traffic /
 school traffic / peak time

*eg mean greater due to rush hour, or λ smaller due to
 congestion, or 2.6 too small due to school traffic*

B1

2

- (c) $Y \sim \text{Bin}(20, 0.2)$
 $P(Y \geq 5) = 1 - P(Y \leq 4)$
or:



$$1 - \left(\frac{0.01153 + 0.05765 + 0.13691}{+0.20536 + 0.21820} \right)$$

$$= 1 - 0.6296$$

$$1 - 0.6296$$

(Allow 1 - 0.8042 seen for M1)

M1

$$= 0.37(0) \text{ (3sf)}$$

AWFW 0.37 to 0.3704

A1

2

(d) X and Y independent

Any statement which indicates two / both events are independent

B1

$$p = 0.0177 \times 0.3704$$

[their (c)] $\times$ [their (a)(ii)]

M1

$$= 0.00656 \text{ (3sf)}$$

AWFW 0.0065 and 0.0067

A1

3

[13]

EXAM PAPERS PRACTICE

Q9.

$X \sim \text{Po}(\lambda)$ ©2025 Exam Papers Practice. All rights reserved.

(a) $E(Z) = 4 \times 2.5 + 30 = 40$

CAO

B1

$$\text{Var}(Z) = 4^2 \times 2.5$$

Use of $V(aX) = a^2 V(X)$

Ignore '+30'

M1

$$= 40 \text{ } \{= E(Z)\}$$

CAO AG

A1

Note:

$$4 \times 2.5 + 30 = 4^2 \times 2.5 \Rightarrow \text{B1 M1 A0}$$

plus value of 40 quoted $\Rightarrow$ B1 M1 A1

3

- (b) No values less than 30 are possible
No odd values are possible
Only even values are possible
Only values of 30, 34, 38, etc, are possible

Or equivalent

B1

1

[4]

Q10.

- (a) X = no. with blood disorder

for $X \sim B(25, 0.7)$

$$P(X > 15) = P(X \geq 16)$$

Alternative:

$$X \sim B(25, 0.7)$$

$$\begin{aligned} P(X > 15) &= 1 - P(X \leq 15) \\ &= 1 - 0.18943 \\ &= 0.81057 \end{aligned}$$

Consider $X' \sim B(25, 0.3)$ then:

$$\begin{aligned} P(X \geq 16) &= P(X' \leq 9) \\ &= 0.8106 \end{aligned}$$

B3 $0.81 \leq p \leq 0.811$

B2 for $0.902 \leq p \leq 0.9022$

B1 for $0.5 \leq p \leq 0.95$

B3, 2, 1

3

©2025 Exam Papers Practice. All rights reserved.

- (b) (i) $X \sim P_0(2.6)$
 $P(X \leq 5) = 0.951$
AWRT

B1

1

- (ii) $Y \sim P_0(4.9)$

$\lambda = 4.9$ stated or used in poisson expression

B1

$$P(Y = 10) = \frac{e^{-4.9} \times (4.9)^{10}}{10!}$$

S

$$= 0.0164$$

A1

3

- (iii) $T \sim P_0(7.5)$
2.6 + (their mean in (ii))

B1ft

$$P(T > 16) = 1 - P(T \leq 16)$$

(for 0.9980)

M1

$$= 1 - 0.9980$$

$$= 0.002$$

CAO (0.00196)

A1

3

[10]

Q11.

- (a) (i) $X \sim \text{Po}(7)$
 $P(X \leq 5) = 0.301$
AWFW 0.300 and 0.301

B1

1

(ii) $P(X = 7) = \frac{e^{-7} \times 7^7}{7!}$

M1

$$= 0.149$$

A1

$$P(X \leq 7) - P(X \leq 6)$$

$$= 0.5987 - 0.4497 \text{ (M1)}$$

$$= 0.149 \text{ (A1)}$$

2

- (iii) $0.65 \leq p \leq 0.66$
 $P(X \leq 9) - P(X \leq 4)$

B3

$$0.72 \leq p \leq 0.73 \quad \text{or} \quad 0.52 \leq p \leq 0.53$$

$$P(X \leq 10) - P(X \leq 4)$$

$$P(X \leq 9) - P(X \leq 5)$$

(B2)

0.60

$$P(X \leq 10) - P(X \leq 5)$$

(B1)

3

- (b) No. telephone calls received per hour
 $= Y \sim P_0(0.875)$

B1

1

- (c) (i) Maximum number = 4

B1

1

(ii) $P(Y < 4) = P(Y = 0, 1, 2, 3)$
 $= e^{-0.875} \left(1 + \frac{7}{8} + \frac{49}{128} + \frac{343}{3072} \right)$
 $= 0.4169(1 + 0.875 + 0.3828 + 0.1117)$
 $= 0.987740443$
Any correct expression (B2)
or AFWW 0.987 to 0.988

B2

$$P(Y \geq 4) = 1 - 0.9877$$

$$1 - (\text{their } P(Y < 4))$$

M1

$$= 0.0123$$

©2025 Exam Papers Practice. All rights reserved.
 AFWW 0.0122 and 0.0123

A1

4

SC

$$P(Y \leq 4) = 0.997 \text{ to } 0.998$$

or any correct expression

B2

$$P(Y > 4) = 0.002 \text{ to } 0.003 \quad \text{M1 A0}$$

- (d) λ probably not constant
 The number of calls in any time interval
 of 1 hour is likely to vary throughout the day.
 'System Down'
 $\Rightarrow$ not independent

E1

1

[13]

Q12.

(a) (i) $P(X \leq 3) = 0.515$
 0.5152

B1

1

(ii) $P(Y = 5) = \frac{e^{-4.4} \times (4.4)^5}{5!}$

$P(Y \leq 5) - P(Y \leq 4) = 0.7199 - 0.5512$
 correct values seen

M1

$= 0.169$
 (0.1687)

A1

2

(b) (i) $T = \text{Po}(8.0)$

B1

X and Y are **independent**
 (Poisson random variables)

B1

2

(ii) $P(6 < T < 12) = P(T \leq 11) - P(T \leq 6)$

M1

$= 0.8881 - 0.3134$

A1

$= 0.575$
 (0.5747)

A1

3

(iii) $P(T > 14) = 1 - P(T \leq 14)$

M1

$= 1 - 0.9827$
 $= 0.0173$
 CAO

A1

$p = (0.0173)^2$

$$[their P(T > 14)]^2$$

M1

$$= 0.0003 \text{ (1 sf)}$$

ft if $0 < \text{both } p\text{'s} < 1$

A1F

4

$$(iv) \quad P(T \leq k) > 0.99 \Rightarrow k \geq 15$$

$$P(T \leq 15) = 0.9918$$

$$P(T \leq 14) = 0.9827$$

M1

$\therefore$ minimum number of devices that
Joe should keep in stock = 15

A1

2

[14]

Q13.

$$(a) \quad (i) \quad X \sim \text{Po}(5.0) \\ \Rightarrow P(X < 4) = P(X \leq 3) \\ (0.440 \text{ to } 0.441) \text{ for B1}$$

$$= 0.265$$

CAO

B2

2

$$(ii) \quad Y \sim \text{Po}(1.5)$$

$$\Rightarrow P(Y = 4) = \frac{e^{-1.5} \times (1.5)^4}{4!}$$

M1

$$= 0.0471$$

(0.047 to 0.0471)

A1

2

$$(b) \quad (i) \quad T = X + Y \sim \text{Po}(6.5)$$

B1

$$\Rightarrow P(T > 5) = 1 - P(T \leq 5)$$

(1 - 0.2237) **or** (1 - 0.5265)

B1

$$= 1 - 0.369$$

$$= 0.631$$

B1

3

(ii) $p = {}^8C_7 (0.631)^7 (0.369) + (0.631)^8$

ft on their p from (b)(i)
Either part attempted

M1ft

$$p = 0.11758 + 0.02513$$

(both parts correct)

A1ft

$$= 0.143$$

AWFW 1.142 to 0.143 (CAO)

A1

3

(c) (i) Mean = 8
 CAO

B1

$$\text{Variance} = s^2 = 16.9$$

(sample variance = 15.2)
(AWRT)

B1

2

©2025 Exam Papers Practice. All rights reserved.

(ii) Poisson not a good model for data

B1dep

$$\text{Mean} \neq \text{Variance}$$

B1

2

[14]

Q14.

(a) (i) $X \sim \text{Po}(9 \ 0) \Rightarrow$
 standard deviation = 3

B1

1

(ii) $P(6 < X < 12)$

$$= P(X \leq 11) - P(X \leq 6)$$

M1

$$= 0.8030 - 0.2068$$

M1ft

$$= 0.5962$$

CAO

A1

3

(b) (i) $T \sim \text{Po}(11.5)$

CAO

B1

1

(ii) $P(T \leq 1) = P(T = 0) + P(T = 1)$
 $= e^{-11.5} + 11.5e^{-11.5}$
 $= 0.000127$

Use of $T = 0$ and 1

M1

Substitute correctly into formula

M1

AWFW 0.000126 and 0.00013

A1

3

(c) $\bar{x} = 12.0$ and $s^2 = 19.3$ ($s = 4.40$)

$$\sigma^2 = 17.4 \text{ } (\sigma = 4.17)$$

B1

Mean and variance very different

dep on s^2 (or σ^2)

E1

$\Rightarrow \text{Po}(12.0)$ not a suitable model

(dep E1)

E1

3

[11]

Q15.

(a) $P(X < 6) = P(X \leq 5)$

M1

$$= 0.369$$

A1

2

(b) $P(Y \geq 1) = 1 - P(Y = 0)$

M1

$$= 1 - e^{-1.5}$$

M1

$$= 1 - 0.223$$

$$= 0.777$$

A1

3

(c) (i) $X + Y \sim \text{Po}(8)$

$$\frac{e^{-8} \times 8^6}{6!}$$

M1

$$P(X + Y = 6) = 0.122$$

$$0.3134 - 0.1912$$

M1A1

3

(ii) $P(Y \geq 1 | X < 6) = P(Y \geq 1) = 0.777$

B1

1

[9]

Q16.

(a) $P(X = 8) = P(X \leq 8) - P(X \leq 7)$

$$= 0.8472 - 0.7440$$

$$P(X = 8) = \frac{e^{-8} (8^8)}{8!}$$

M1

$$= 0.103$$

A1

2

(b) (i) $\lambda = 9$

B1
1

(ii) $P(Y > 9) = 1 - P(Y \leq 9)$
 $= 1 - 0.5874$

M1

$= 0.4126$
AWFW 0.412 to 0.413

A1ft
2

(c) (i) $T \sim \text{Po}(15)$

B1ft
1

(ii) $P(T \leq 20) = 0.917$

B1ft
1

(iii) $P(T \text{ at least } 21) = 0.083$

B1ft

$p = 15 \times (0.083)^4 (0.917)^2$
B(6, (iii)) used

$= 0.000599$

CAO; AWW 0.000598 to 0.0006

M1
A1
3

[10]

Q17.

(a) (i) $X \sim \text{Po}(0.70)$
 $P(X < 3) = P(X \leq 2)$
 $= 0.966$
0.9659

B1
1

(ii) $Y \sim \text{Po}(1.30)$

$$P(Y = 2) = \frac{e^{-1.30} (1.30)^2}{2!} = 0.230$$

M1A1

2

(b) $T \sim \text{Po}(2.0)$

M1

$$P(T \geq 4) = 1 - P(T \leq 3) \\ = 1 - 0.8571$$

M1

$$= 0.143$$

$$0.1429$$

A1

3

[6]

Q18.

(a) (i) $P(A = 4) = \frac{e^{-3.5} \times (3.5)^4}{4!} = 0.189$

M1A1

2

(ii) $P(B \leq 6) = 0.762$

B1

1

(iii) $T = A + B \sim \text{Po}(8.5)$

$$P(T \text{ fewer than } 10) = P(T < 10) \\ \text{Use of Po}(8.5)$$

M1

$$= P(T \leq 9)$$

$$T \leq 9 \text{ attempted}$$

M1

$$= 0.653$$

$$\text{CAO}$$

A1

3

(b) $X \sim B(5, 0.653)$

$$X \sim B(5, \text{their } p)$$

B1

$$P(X \geq 4) = \binom{5}{4} (0.653)^4 (0.347) + (0.653)^5$$

M1

$$\begin{aligned} &= 0.31547 + 0.11873 \\ &= 0.434 \end{aligned}$$

On their p from (a)(iii)

A1ft

3

(c) (i) $\bar{x} = 9.2$

B1

$$s^2 = 9.29$$

$$\sigma^2 = 8.36$$

B1

2

- (ii) Mean and variance have similar values which suggests that Poisson distribution may be appropriate

B1ftB1ft

2

EXAM PAPERS PRACTICE [13]

Q19.

©2025 Exam Papers Practice. All rights reserved.

(a) (i) $P(X = 3) = \frac{e^{-3.5} \times (3.5)^3}{3!} = 0.216$

M1A1

2

(ii) $P(Y \geq 5) = 1 - P(Y \leq 4)$
used

M1

$$\begin{aligned} &= 1 - 0.2851 \\ &= 0.715 \end{aligned}$$

A1

2

(b) (i) $T \sim \text{Po}(9.5)$

B1

1

$$(ii) \quad P(7 \leq T \leq 10) = P(T \leq 10) - P(T \leq 6)$$

M1

$$= 0.6453 - 0.1649$$

A1

$$= 0.408$$

Accept 0.48

A1

3

$$(iii) \quad p = (0.4804)^3 = 0.111$$

M1A1ft

2

[10]

Q20.

$$(a) \quad (i) \quad P(X \leq 1) = 0.9098$$

M1

AWRT 0.91

A1

2

$$(ii) \quad P(Y = 2) = \frac{e^{-2.5} \times (2.5)^2}{2!}$$

M1

$$= 0.257$$

AWFW 0.256 and 0.257

A1

2

$$(b) \quad (i) \quad T = X + Y \sim P_o(3.0)$$

$$P(\text{Total of 2 breakdowns}) = p([x, y] = 2)$$

$$= P([2, 0] \text{ or } [0, 2] \text{ or } [1, 1])$$

B1

$$P(T = 2) = \frac{e^{-3.0} \times (3.0)^2}{2!}$$

$$= [0.0758 \times 0.0821] + [0.6065 \times 0.25652]$$

$$+ [0.20521 \times 0.3033] = 0.006223 + 0.155585$$

M1

$$\begin{aligned}
 &= 0.224 \\
 &\quad + 0.062233 \\
 &= 0.224
 \end{aligned}$$

A1

3

OR

$$\begin{aligned}
 \text{Using tables: } &= 0.4232 - 0.1991 \\
 &= 0.224
 \end{aligned}$$

(ii) $p = (0.224)^4 = 0.0025$

M1 A1

2

(c) (i) $4X \sim P_o(2.0) \Rightarrow \mu = 12.0$
 $4Y \sim P_o(10.0)$

B1

1

(ii) $P(T \geq 18) = 1 - P(T \leq 17)$

M1

$$\begin{aligned}
 &= 1 - 0.9370 \\
 &= 0.0630
 \end{aligned}$$

A1

2

[12]

EXAM PAPERS PRACTICE

Q21.

©2025 Exam Papers Practice. All rights reserved.

(a) (i) $P(X = 2) = \frac{e^{-1.5} \times (1.5)^2}{2!} = 0.251$

M1A1

2

(ii) $p = (0.251)^3 = 0.0158$
on their p from (i)

M1A1ft

2

(b) (i) $Y \sim P_o(9.0)$

B1

1

(ii) $P(Y \geq 12) = 1 - P(Y \leq 11)$
 $= 1 - 0.8030$

M1

$$= 0.197$$

A1

2

- (c) attacks patients:
randomly (p constant)
 $\text{mean of } 1.5 \Rightarrow p \text{ small (B1)}$
(unless very few patients)

B1

independently

B1

[9]

Q22.

- (a) For a 1-year period
The number of A grades $\sim \text{Po}(3)$

For a 5-year period
The number of A grades $\sim \text{Po}(15)$

B1

$$\begin{aligned} P(\text{Total A-grades} > 18) \\ = 1 - (\text{Total} \leq 18) \end{aligned}$$

M1

$$\begin{aligned} &= 1 - 0.8195 \\ &= 0.1805 \\ &= 0.181 \end{aligned}$$

AWFW 0.180 and 0.181

A1

3

- (b) (i) $X + Y \sim \text{Po}(10)$

B1

$$\begin{aligned} P(X + Y \leq 14) &= 0.917 \\ &(0.916 \text{ to } 0.917) \end{aligned}$$

M1A1

3

- (ii) X and Y are independent variables

E1

1

[7]

Q23.

(a) (i) $P(Y = 2) = \frac{e^{-1.9} \times (1.9)^2}{2!}$

M1

$= 0.270$

AWRT

A1

2

(ii) $(0.270)^5 = 0.00143$

On their (a)(i)

AWRT

M1A1ft

2

(b) (i) $X \sim P_o(9.5)$
Poisson and 9.5

B1

1

(ii) $P(X \geq 10) = 1 - P(X \leq 9)$

M1

$= 1 - 0.5218$

$= 0.4782$

A1

2

(iii) $\therefore p = 10 \times (0.4782)^3 (0.5218)^2$

On their b (ii)

M1

$= 0.298$

AWRT 0.3

A1

2

[9]

Q24.

(a) $P(X = 2) = \frac{e^{-2.6} (2.6)^2}{2!}$

$0.5184 - 0.2674 = 0.251$

M1

$$= 0.251$$

A1

2

- (b) (i) $Y \sim P_0(13)$
Poisson and 13

B1

1

(ii) $P(Y \geq 15) = 1 - P(Y < 14)$

M1

$$\begin{aligned}
 &= 1 - 0.6751 \\
 &= 0.3249 \\
 &= 0.325
 \end{aligned}$$

On their λ from b (i)

A1ft

$$\therefore p = (0.3249)^4$$

M1

$$p = 0.0111 \text{ to } 0.0112$$

On their $p(Y \geq 15)$

A1ft

4

EXAM PAPERS PRACTICE [7]

Q25.

©2025 Exam Papers Practice. All rights reserved.

- (a) (i) $X \sim P_0(4.0)$

$$P(X > 8) = 1 - P(X \leq 8)$$

M1

$$\begin{aligned}
 &= 1 - 0.9786 \\
 &= 0.214
 \end{aligned}$$

(0.021) accept

A1

2

- (ii) $X \sim P_0(3.5)$
 $P(Y < 2) = e^{-3.5}(1 + 3.5)$

M1

$$= 0.136$$

(0.13589)

A1

2

(b) (i) $\lambda = E(T) = 7.5$

B1

1

(ii) $P(T \geq 11) = 1 - P(T \leq 10)$

M1

$$= 1 - 0.8622$$

$$= 0.1378$$

(on their λ)

A1f.t.

2

[7]

Q26.

X = no. of strikes per game

$$\therefore X \sim P_0(\lambda)$$

$$X_1 \sim P_0(0.2)$$

$$X_2 \sim P_0(1.1)$$

$$X_3 \sim P_0(1.1)$$

$$\therefore T = X_1 + X_2 + X_3 \sim P_0(2.4)$$

Adding 3 values must be $P_0(2.4)$

M1A1

$$P(3 \leq T \leq 5) = P(T = 3 \text{ or } 4 \text{ or } 5)$$

Seen/attempt for $P_0(\lambda)$ only

©2025 Exam Papers Practice. All rights reserved.

M1

$$= P(X \leq 5) - P(X \leq 2)$$

(Poisson tables only)

M1

$$= 0.9643 - 0.5697$$

$$= 0.3946$$

AWRT 0.395

A1

5

[5]

Q27.

- (a) (i) $\lambda = 3.6$ for a 5-minute period
let Y = no. of vehicles arriving
in a 5 minute period

then $Y \sim P_o(3.6)$

$$P(Y \leq 2) = 0.3027$$

$$P(Y \geq 3) = 1 - P(Y \leq 2)$$

0.303 (AWRT)

Use of tables or $P(Y = 0) = e^{-3.6} = 0.2732$

B1/M1

$$= 1 - 0.3027$$

$$P(Y = 1) = 3.6 e^{-3.6} = 0.09837$$

$$= 0.6973$$

$$= 0.697$$

$$P(Y=2) = \frac{(3.6)^2 e^{-3.6}}{2!} = 0.17706$$

A1

3

(ii) $p = 0.6973$ (or 0.697)

use of their $P(Y \geq 3)$

M1

P(at least 3 arrive in 3 succ.5 m)

$$= (0.6973)^3$$

p^3 , iff first M1 gained

m1

$$= 0.339046$$

$$= 0.339$$

CAO

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

A1

3

(b) Let $X = \#$ vehicles arriving in a 10 - minute period

then $X \sim P_o(7.2)$

FOR 7.2

B1

$$P(X = 0) = e^{-7.2}$$

$$= 0.0007$$

$$\left\{ \begin{array}{l} e^{-3.6} \times e^{-3.6} \\ \text{for Poisson, correctly} \\ AG(0.0007463) \end{array} \right.$$

M1A1

3

[9]

Q28.

- (a) (i) $P(2 \text{ or fewer}) = 0.9526$
 0.953 (0.952 ~ 0.953)

B1

- (ii) $P(>1, <3) = P(2)$
 P(2) required

M1

$= P(2 \text{ or fewer}) - P(1 \text{ or fewer})$
 P(2 or fewer) – P(1 or fewer) or correct use of formula

m1

$= 0.9526 - 0.8088 = 0.144$
 0.144 (0.143 ~ 0.1445)

A1

4

- (b) (i) mean $5 \times 0.8 = 4$
 4 cao

B1

standard deviation $\sqrt{4} = 2$

M1

$\sqrt{\text{their mean} - \text{allow variance} = \text{their mean} - \text{variance must be stated}}$
 2 cao

A1

3

- (ii) $P(\text{fewer than } 3) = 0.2381$
 (0.2375 – 0.2385)

B1

1

[8]

Q29.

- (a) Poisson
 cao

B1

1

- (b) (i) $P(2 \text{ or fewer}) = 0.570$
 0.570 (0.569 to 0.57)

B1

- (ii) $P(4) = 0.9041 - 0.7787$
 $P(4) = P(4 \text{ or fewer}) - P(3 \text{ or fewer})$ or correct
 use of Poisson formula

M1

$= 0.125$
 $0.125 (0.125 \text{ to } 0.126)$

A1

3

- (c) (i) Poisson, mean 12
Poisson, mean 5×2.4 (may be implied)

M1

Poisson mean 12 (may be implied)

A1

$P(\text{fewer than } 6) = 0.0203$
 $0.0203 (0.02 \text{ to } 0.021)$

B1

- (ii) $P(> 17) = 1 - 0.9370 = 0.0630$
 $P(>17) = 1 - P(17 \text{ or fewer})$

M1

$0.063 (0.0625 \text{ to } 0.0635)$

A1

5

EXAM PAPERS PRACTICE [9]

©2025 Exam Papers Practice. All rights reserved.

Q30.

- (a) (i) 0.9526
 $0.9526 (0.952 - 0.953)$

B1

1

- (ii) $1 - 0.4493 = 0.5507$
 $P(\text{at least once}) = 1 - P(0)$ or use of correct formula
 $0.5507 (0.55 - 0.551)$

M1 A1

2

- (iii) $0.8088 - 0.4493 = 0.3595$
 $P(1) = P(1 \text{ or fewer}) - P(0)$ or use of correct formula
 $0.3595 (0.359 - 0.36)$

M1 A1

2

- (b) (i) Poisson mean 4
Po(4) may be implied later but not if answered incorrectly

B1
1

- (ii) $1 - 0.6288 = 0.3712$
 $1 - P(4 \text{ or fewer})$
 $0.3712(0.371 - 0.372)$

M1 A1
2

- (iii) $0.5507^5 = 0.0506$
Use of their (a)(ii)
[their P(at least once)]⁵
 $0.0506(0.0506 - 0.051)$

M1 M1 A1
3

[11]

Q31.

- (a) 0.977

Correct use of tables/formula
 $0.977(0.976 - 0.977)$

M1/A1
2

- (b) Poisson mean 3
Mean 3

B1

©2025 Exam Papers Practice. All rights reserved.
 $P(>4) = 1 - 8153$
Method, allow 1 - 3 or fewer

M1

$= 0.185$
 $0.185 (0.185 - 0.186)$

A1
3

- (c) Individuals do not enter independently
E2(1) independent

E2(1)
2

[7]

Q32.

(a) (i) 0.231
 $0.23 - 0.232$

B1
1

(ii) $1 - 0.4695$
 $P(3 \text{ or more}) = 1 - P(2 \text{ or fewer})$ or
correct method of calculation

M1

$= 0.5305$
 $0.53 - 0.531$

A1
2

(b) Poisson mean 5×2.8
completely correct method

M1 M1

$P(<10) = 0.109$
 $0.109 - 0.11$

A1
3

(c) (i) $x = 1.3$
cao

B1

$S^2 = 3.34$
 $3.34 - 3.35$
allow 3.01 (3.005 - 3.015)

B1
2

(ii) Mean and variance different

B1
1

(iii) Mean unlikely to be constant over 24 hour period/not independent

B1
1

[10]

Q33.

(a) (i) $P(2 \text{ or fewer}) = 0.8335$
 $0.8335(0.833 - 0.834)$

B1

- (ii) Poisson mean 2.8
Poisson mean 2×1.4

M1

$$P(4 \text{ or more}) = 1 - 0.6919$$

$$P(4 \text{ or more}) = 1 - P(3 \text{ or fewer})$$

Method

M1 M1

$$= 0.308$$

A1

5

$$0.308(0.3075 - 0.3085)$$

- (b) Poisson mean 2.4

M1

$$P(<4) = 0.7787$$

A1

2

$$0.779(0.778 - 0.779)$$

- (c) Poisson mean 3.8

$$\text{Poisson mean } 1.4 + 2.4$$

M1

$$P(> 6) = 1 - 0.9091$$

$$P(> 6) = 1 - P(6 \text{ or fewer})$$

m1

$$= 0.0909$$

$$0.0909(0.908 - 0.091)$$

A1

3

- (d) Not independent/ mean may vary according to day of the week

E1

1

[11]

Examiner reports

Q2.

This question was generally very well-answered, although a small proportion used incorrect values of λ and scored poorly. In the final part, some students merely used 'standard' Poisson phrases such as 'not independent' or 'mean not equal to variance', but most 'considered the context and produced a sensible explanation.

Q3.

Part (a)(i) was generally well answered and, in parts (ii) and (iii), the value of λ was usually correctly calculated, but the boundary values to be used caused more difficulty. In part (b), answers such as "The days are independent" or "The satellites are independent" showed why it could not be assumed that "They are independent" referred to the GRBs. A not uncommon answer was "Because mean and variance are equal" which showed that students were quoting learnt answers, without any regard for the information given, or not given, in the question.

Q4.

A small minority of students used the normal approximation to the Poisson in part (a) and so scored minimal marks. The majority, who used Poisson tables or their calculators' equivalent, often scored at least 3 marks with the usual mark lost for quoting 21 or 23 instead of 22 in (b).

Q5.

In part (a)(i), the vast majority of candidates correctly stated the distribution of X to be 'Poisson'. In part (a)(ii), although most stated the answers correctly as $E(3X - 1) = 3\lambda$ and $\text{Var}(3X - 1) = 9\lambda$ some, who seemed to understand the underlying concepts by stating $E(3X - 1) = 3 \times E(X) - 1$ and $\text{Var}(3X - 1) = 3^2 \times \text{Var}(X)$, then failed to state their answers in terms of λ as requested, consequently gaining no marks. Part (a)(iii) proved to be very

popular with most candidates. They were able to correctly state $P(X = x + 1) = \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!}$ and then were able to use

$\lambda^{x+1} = \lambda \times \lambda^x$ and $(x + 1)! = (x + 1) \times x!$ to prove the given answer.

In part (b), it was very pleasing to see that candidates dealt with such phrases as 'at least 10' and 'at most 3' much better than in previous series. In part (b)(i), $\lambda = 8.5$ followed by a correct calculation was often seen. In part (b)(ii), $\lambda = 14.3$ was correctly seen and often used to find the correct answer of 0.00037. Unfortunately, some failed to comply with the instruction 'give your answer to two significant figures' and consequently lost a mark. Those who chose to do the whole of this part of the question on their calculators either gained full marks when the correct answer was stated or lost at least two marks for incorrect answers with no method shown.

Q6.

Many fully successful solutions were seen to part (a). In part (b), some students were unable to cope with the fact that the value of $\lambda = 1.5$ was not given in Table 2 of the

supplied booklet. Thus some tried to use this table by finding the average of the values of $P(X \leq 1)$ for $\lambda = 1.4$ and $\lambda = 1.6$. This incorrect method gained no credit. Answers to part (c)(i) were almost always correct as were those to part (c)(ii), except for the students who thought that they needed to evaluate $1 - P(X \leq 15)$. The great majority of students realised that the binomial distribution was required in part (c)(iii). Unfortunately, one of the two required terms was often missing.

Q7.

Part (a)(i) was answered very well. However, in part (a)(ii), some candidates attempted to use $Po(1.5)$ or, because $Po(0.15)$ was not in the booklet provided, attempted to average the values found under $Po(0.10)$ and $Po(0.20)$. Neither of these approaches gained any credit. In part (b), it was expected that candidates would state either that ' X and Y are independent' or that 'the number of catches and run-outs are independent'. Although these were often stated, there were candidates who thought that stating 'they are independent' or that 'cricket matches are independent' would suffice. These statements were not sufficient to gain any credit.

Parts (c)(i) and (c)(ii) were usually very well answered. However, some candidates seemed to have some difficulty interpreting statements such as 'at least 15'.

Q8.

In part (a)(i), candidates were expected not only to give the value of λ to be 13 but also to state that X followed a Poisson distribution; this was not always seen. Part (a)(ii) was usually done well either by using tables or by applying the correct formula. Unfortunately some candidates used $\lambda = 2.6$. In part (a)(iii), tables were often used correctly and accurately to obtain the correct answer of 0.920. Part (b) was not done at all well, with only the most able candidates gaining any credit. Although many thought that the value of λ would be different from 2.6, they could not give a convincing reason why this should be the case.

Comments such as 'It is not random' or 'Mean not equal to the variance' by themselves were not specific enough and consequently gained no credit. Part (c) was usually very well answered by the vast majority of candidates who used tables. Those who chose to use the formula approach usually either missed one of the terms out or made some error in their calculations. In part (d), the necessary assumption that was expected was ' X and Y are independent'. Comments such as 'They are independent' or 'The speeds are independent' gained no credit. The calculation of the required probability was usually attempted correctly.

Q9.

Most candidates provided a convincing answer in part (b). Also, it was very rare to see an incorrect answer to part (a) but even rarer to see a correct reason in part (b), where the most common answer was ' λ too large', which, needless to say, did not score the mark.

Q10.

Only the most able candidates managed to find the solution to part (a) by stating correctly that $P(X > 15) = 1 - P(X \leq 15) = 1 - 0.18943 = 0.811$. For those using tables for $B(25, 0.3)$, the most popular misconceptions were to obtain $P(X \leq 15) = 0.9995$, and then **either** thinking that this gave the correct answer **or** that, since the value of $0.3 p - =$ had been obtained by subtracting 0.7 from 1, the correct answer was found by subtracting 0.9995

from 1. Very few candidates were able to use the binomial tables correctly and so $P(X > 15) = P(X' \leq 9) = 0.8106$ was very rarely seen.

Answers to part(b)(i) were usually correct as were those to part (b)(ii), except by those candidates who, on not finding $\lambda = 4.9$ in the tables, incorrectly thought that the use of $\lambda = 5$ was close enough. In part (b)(iii), candidates were asked to write down the **distribution** of the random variable T . Although the majority stated that $\lambda = 7.5$ they did not state that the distribution was Poisson and consequently lost a mark. There were many fully correct solutions to $P(T > 16)$ with most candidates realising that $P(T > 16) = 1 - P(X \leq 16) = 0.002$.

Q11.

The majority of candidates gained full marks for parts (a)(i) and (a)(ii). However, in part (a)(iii), although there were many correct answers seen, there were far too many candidates who could not interpret correctly the expression 'at least 5 but fewer than 10'.

In part (b), candidates were asked to write down the **distribution** of the number of telephone calls received each hour. Although the majority stated that $\lambda = 0.875$, they did not state that the distribution was Poisson and consequently lost a mark. Part (c)(ii) was answered correctly by, in the main, only the most able candidates. Some who correctly realised that $P(Y \geq 4) = 1 - P(Y \leq 3)$ was required then omitted $P(Y = 0)$ when evaluating $P(Y \leq 3)$ or used $\lambda = 7$ in their calculations. The reasons given in part (d) were not always expressed well enough to gain credit.

Q12.

This was another question which provided many candidates with a good source of marks. Most were able to gain the correct answers to part (a). In part (b)(i), the great majority of candidates realised that **both** Poisson and $\lambda = 8$ were required. When stating their assumption, some candidates simply stated 'they are independent' without any actual indication of *what* was independent; this did not gain any credit. In part (b)(ii), whilst most candidates realised that $P(6 < X < 12) = P(X = 7, 8, 9, 10, 11)$ and consequently went on to evaluate $P(X \leq 11) - P(X \leq 6)$ using the tables provided to gain the correct answer, some seemed less certain as to what values were included in $6 < X < 12$. In part (b)(iii), the great majority of candidates correctly translated 'exceed 14' and so attempted to find $P(T > 14)$ by evaluating correctly $1 - P(T \leq 14)$. This was invariably followed correctly by $p = 2 \times P(T > 14)$ (1sf).

Unfortunately some candidates thought that $P(T > 14) = 1 - P(T \leq 13)$ and that $p = 2 \times P(T > 14)$. Candidates should realise that where a required degree of accuracy is stated in the question, then they must adhere to this request. Many failed to do this, giving their answer as 0.000299, and consequently lost some credit. In part (b)(iv), there were many good solutions seen with the correct answer of 15 often quoted.

Q13.

There were many fully-correct solutions seen to this question. The vast majority of candidates used the tables provided to enable them to find the correct answer to part (a)(i) by the most efficient method. A minority of candidates seemed only able to use the formula and consequently worked out the individual probabilities before obtaining their answer. In part (a)(ii), the formula was often used correctly to obtain the correct answer of 0.0471. However, there were some candidates who seemed to think that they had to use tables whatever the value of λ and so attempted to average the values found in the tables

under $\lambda = 1.4$ and $\lambda = 1.6$ incorrectly thinking that this would give them the required value for $\lambda = 1.5$.

In part (b)(i), most candidates interpreted ‘more than 5’ correctly using $P(X + Y > 5) = 1 - P(X + Y \leq 5)$, whilst others incorrectly used $1 - P(X + Y \leq 4)$ or $1 - P(X + Y \leq 6)$. Only the most able candidates then went on to use their answer to correctly determine the numerical solution to part (b)(ii). There were many erroneous attempts at using various expressions with $0.631^7 + 0.631^8$ or $0.631^7 \times 0.369$ or even $6.5 \times 0.631^7 \times 0.369$ seen, rather than the required $p = {}^8C_7(0.631)^7(0.369) + (0.631)^8$. In part (c)(i), the vast majority of candidates correctly stated that $\bar{x} = 8$ and that $s^2 = 16.9$, but with some stating the sample variance as 15.2. These answers usually enabled candidates to indicate correctly in part (c)(ii) that a Poisson distribution was probably not a good model since the mean and the variance had very different values.

Q14.

In part (a)(i), there were surprisingly many candidates who did not know the relationship between the value of λ and the standard deviation of a Poisson distribution. In part (a)(ii), although the vast majority realised that the most efficient method of answering this part of the question required the use of the table on page 23 of the formulae booklet, very many failed to look up the appropriate values to enable them to show that $P(6 < X < 12) = P(X \leq 11) - P(X \leq 6) = 0.596$. Those candidates who calculated the separate probabilities for $X = 7, 8, 9, 10$ and 11 often reached the required answer.

In part (b)(i), it should be noted that $\lambda = 11.5$ and an indication that the distribution of T is Poisson were both required to gain the mark. In part (b)(ii), most candidates realised that “fewer than 2” implied that $P(T \leq 1) = P(T = 0) + P(T = 1)$. The correct answer of 0.000127

was then usually found by the use of $\lambda = 11.5$ and the formula $P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$ (found on page 11 of the formulae booklet).

Although part (c) asked candidates to calculate appropriate numerical measures (plural), this was ignored by some candidates. Such candidates usually calculated $\bar{x}$ and, finding this to coincide with the given value of λ , stated inappropriately that Po(12) was a suitable model for the given data. Of those candidates who did calculate more than one numerical measure, most correctly calculated the mean and variance of the given data ($\bar{x} = 12$; $s^2 = 19.3$ or $\sigma^2 = 17.4$), and then indicated that, because the mean and variance were (very) different values, Po(12) was not a suitable model for the number of places booked each week. Some candidates incorrectly thought that the mean and the standard deviation had to be the same.

Q15.

The majority of candidates managed to complete correctly parts (a), (b) and (c)(i) but some could only give the correct answer to part (a), apparently finding the remainder of the question beyond them. In part (c)(ii), all candidates failed to realise that $P(Y \geq 1 | X < 6) = P(Y \geq 1)$, and consequently no correct answers were seen.

Q16.

Part (a) was answered well by most candidates. The use of tables to evaluate $P(X = 8) = \frac{e^{-6} \times 6^8}{8!}$ or the use of the formula $P(X = 8) = \frac{e^{-6} \times 6^8}{8!}$ were equally successful

methods used here. In part (b)(i), although the correct answer of $\lambda = 9$ was often seen, it was common to see the wrong answer of $\lambda = 3$. In part (b)(ii), the most prevalent error was made by candidates who wrongly thought that $P(Y > 9) = 1 - P(Y \leq 8)$. In part (c)(i), where candidates were asked for the distribution, it was not sufficient to simply quote the value of λ without also indicating that the distribution was Poisson. In part (c)(ii), the phrase ‘at most 20 telephone calls’ was sometimes misinterpreted, with candidates wrongly attempting to evaluate $P(T > 20)$ instead of $P(T \leq 20) = 0.917$. In part (c)(iii), many candidates did not realise that $P(\text{at least } 21) = 1 - P(\text{at most } 20) = 0.083$. Even when this was realised, many candidates then thought that the required answer could be found by evaluating either 0.083^4 or $(0.083)^4 \times (0.917)^2$. This gave scant regard to the fact that exactly 4 one-hour periods can be achieved from the given 6 one-hour periods in

$$\binom{6}{4} = 15$$

different ways. It was expected that $B(6, 0.083)$ would be used here; this was often not the case.

Q17.

This question was very well answered, with the majority of candidates gaining full marks. In part (a)(i), some candidates quoted $P(X < 3) = P(X \leq 2)$ and then used the tables incorrectly, often looking up $P(X \leq 3)$, presumably by mistake. Part (a)(ii) was done well by most candidates. The interpretation, in part (b), of “at least 4” was much better understood than in previous papers although there were still a few candidates who found this difficult.

Q18.

Part (a) was generally answered well. However, there were some candidates who attempted to use tables in part (a)(i), even though values are not tabulated for $\lambda = 3.5$, whilst in part (a)(iii) some candidates could not interpret ‘fewer than 10’ to mean ‘ ≤ 9 ’. It was, however, very pleasing to see fewer candidates than usual attempting to use the formula in part (a)(ii). Only the most able candidates managed to complete part (b) successfully, with the majority not realising that there were 5 ways of selecting 4 days from 5.

Part (c) was done really well by most candidates, although, in part (c)(ii), several thought that the mean had to be equal to the variance in order for a Poisson distribution to form a suitable model. Here, $\bar{x} = 9.2$ and $s^2 = 9.29$ were close enough to each other to draw this conclusion.

Q19.

In part (a)(i), the vast majority of candidates realised that the formula or their calculator had to be used in order to obtain the answer. However, there were some who tried to use Table 2 in the booklet provided by averaging the values that they found under $\lambda = 3.4$ and $\lambda = 3.6$, presumably under the misapprehension that this would give them the required value for $\lambda = 3.5$. However, most candidates did use Table 2 correctly in part (a)(ii) by realising that the most efficient way to obtain the result was by using $P(Y \geq 5) = 1 - P(Y \leq 4)$. Some candidates did not seem to be able to use these tables (or perhaps even know of their existence) as they worked out $1 - P(Y = 0, 1, 2, 3 \text{ or } 4)$ by using the formula. Although they usually managed to arrive at the correct answer, this is not the most efficient way of tackling this type of question.

In part (b)(i), candidates had to indicate that the distribution was Poisson and that $\lambda = 9.5$. It was not sufficient to simply write “ $\lambda = 9.5$ ” or just to state “Poisson”. Part (b)(ii) caused

most of the problems in this question. Whilst the vast majority of candidates realised correctly that $P(7 \leq T \leq 10) = P(T \leq 10) - P(T \leq 6)$, there were some who incorrectly used $P(7 \leq T \leq 10) = P(T \leq 11) - P(T \leq 6)$ or who either used the wrong value of λ or were simply careless and misread the tables. The vast majority of candidates realised that, in part (b)(iii), the correct answer could be obtained by simply evaluating (their part (b)(ii))³. However, there were a few candidates who incorrectly thought that $3 \times$ (their part (b)(ii)) was the way forward and did not seem at all deterred when this gave them an answer greater than one.

Q20.

Answers to part (a)(i), were invariably correct with most candidates using the statistical tables provided. A minority of candidates calculated $P(X = 0) + P(X = 1)$ using the correct probability function for $Po(0.5)$ and were often successful. Part (a)(ii) was done well by all but the weaker candidates who usually attempted to use tables to find $P(X = 2)$ for $\lambda = 2.4$ and $\lambda = 2.6$ and then find the mean of the two values obtained! In part (b)(i), most candidates considered $Po(3.0)$ and then used the formula or tables to achieve correctly the given answer of 0.224. In part (b)(ii) most candidates correctly considered $(0.224)^4$ and so scored the 2 marks available. The majority of candidates, in part(c)(ii), correctly interpreted 'T at least 18' as meaning ' $1 - (T \leq 17)$ '.

Q21.

Part (a)(i) was answered successfully by all but the weakest candidates, with most obtaining the correct value of 0.251. Unfortunately some failed to write their answer to 3 significant figures. In part (a)(ii), too many candidates considered $Po(4.5)$ instead of evaluating $(0.251)^3$ to obtain the correct answer of 0.0158. Most candidates realised that in order to achieve credit in part (b)(i), they had to state both the distribution (Poisson) **and** the value of its parameter ($\lambda = 9.0$). There were many excellent solutions seen to part (b)(ii) with most candidates realising that to find 'P(at least 12)' implies that $P(Y \geq 12) = 1 - P(Y \leq 11)$ has to be considered. As on previous papers, candidates found it very difficult to answer the type of question posed in part (c). Many attempts did not relate to the context used in the question but simply stated general conditions under which a Poisson distribution may be appropriate.

©2025 Exam Papers Practice. All rights reserved.

Q22.

In part (a), the vast majority of candidates were able to state that $A = 15$, with many going on to obtain full credit for the correct answer of 0.181. Candidates still had difficulty with the interpretation of 'more than' with the result that $P(X > 18)$ or $P(X > 18) = 1 - P(X < 17)$ was often seen. In part (b)(i), several candidates used $A = 7$ instead of combining the two Poisson variables X and Y to obtain $X + Y \sim Po(10)$.

Interpretation of 'fewer than 15' was usually stated correctly as ' < 14 ', with tables then used to obtain the required answer 0.917. In part (b)(ii), the idea of 'independence' was usually well understood although some candidates stated incorrectly that the variable was normal.

Q23.

Part (a) was usually answered well. In part (b), it was evident that most candidates realised that a description of the distribution and the value of the parameter were **both** required for full credit. The interpretation of 'at least 10' as ' ≥ 10 ' was usually correctly seen in part (b)(ii), with

$P(X \geq 10) = 1 - P(X \leq 9)$ usually seen and Table 2 in the supplied booklet used correctly to achieve the required answer of 0.478. Part (b)(iii) was found difficult and beyond the capabilities of most candidates.

Q24.

There were many fully correct solutions to part (a) with the vast majority of these being found by the use of Table 2 in the supplied booklet. In part (b)(i), candidates realised that both a description of the distribution (Poisson) and the value of the parameter ($\lambda = 13$) were required. The first step of part (b)(ii) was answered well by most candidates, with the correct interpretation of 'at least 15' being used to give $P(X \geq 15) = 1 - P(X \leq 14) = 0.325$. Unfortunately, some candidates interpreted the phrase 'at least 15' as meaning the same as 'less than 15'. A surprising number of scripts from good candidates were seen where the solution to the second step (four successive) was missing altogether. This was the main source of error on this question.

Q25.

This question proved to be a very good source of marks with many fully correct solutions presented. In parts (a)(i) and (b)(ii), there were very many excellent solutions making good use of tables. However, some candidates used formulae to evaluate probabilities thus losing valuable time that could have been better spent on more demanding parts of the paper. Some candidates were unable to translate 'more than 8', 'fewer than 2' and 'at least eleven' into mathematically equivalent ranges.

Q26.

There were many excellent solutions to this question by candidates who realised that the total number of strikes, Y , could be represented by the distribution $P_0(2.4)$. However, it was wrongly thought by some candidates that $P(3 \leq Y \leq 5) = P(Y \leq 5) - P(Y \leq 3)$ which inevitably led to a loss of marks. Several candidates tried to work separately with $P_0(1.1)$ $P_0(0.2)$ and consequently found this straightforward question very difficult.

Q27.

This question was well answered with many students gaining full marks.

In part (a)(i) the vast majority of candidates realised that $P(Y \geq 3)$ was required, with most attempting to evaluate $1 - P(Y \leq 2)$. Again, candidates from some centres wasted time by evaluating individual probabilities instead of using the cumulative Poisson probability tables supplied.

In part (a)(ii) most candidates realised that they had to use the value found in part (a)(i), with most evaluating 0.6973^3 to give the correct response of 0.339. Unfortunately, some candidates attempted to use $P_0(10.8)$ and gained no credit.

The majority of candidates managed to gain full marks in part (b) of this question. Most used $P_0(7.2)$ to evaluate $e^{-7.2}$ whilst others correctly used $P_0(3.6)$ to evaluate $e^{-3.6} \times e^{-3.6}$, giving the required answer. It must be emphasised that in this question all working had to be shown since the answer was given. Quoting 0.0007 followed by 'using tables' or 'using calculator' was insufficient to gain any credit in this question.

Q28.

Nearly all candidates were able to use Poisson tables but fewer realised that the

probability of 'more than one but fewer than three' is $P(2)$. Part (b)(i) was generally well answered with fewer than expected confusing standard deviation and variance.

Q29.

Nearly all candidates were able to identify the Poisson distribution and found this an easy starter. A few failed to use Poisson, mean 12, in part (b).

Q30.

Part (a) was well answered. Many found part (b)(i) puzzling, giving answers such as "Normal" before using the correct distribution in part (b)(ii).

Q31.

The numerical parts were usually answered correctly but many candidates were unable to provide a convincing answer to part (c).

Q32.

This question was generally well answered. Some candidates in part (c)(i), instead of calculating the variance, assumed it was equal to the mean and so were unable to answer part (c)(ii).

Q33.

Using tables to find Poisson probabilities posed few problems. However, not all candidates used the appropriate mean when Poisson distributions were combined.